

CHEMICAL REACTOR DIGITAL TWIN

Measured silica kinetics benchmark | Open source MIT | vicena.ai

Batch, CSTR, PFR

Mass + energy balances

61 measured experiments

Time-split validation

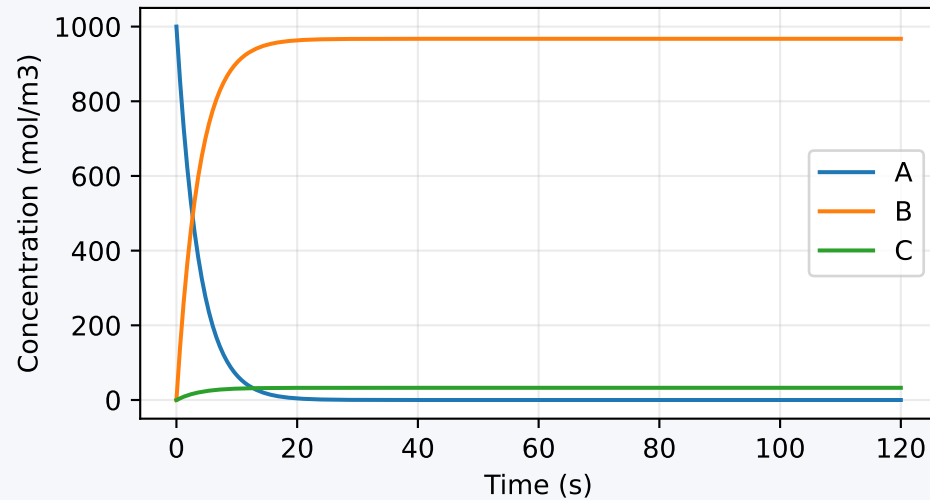
Fourth vs first order

44.42% median RMSE gain

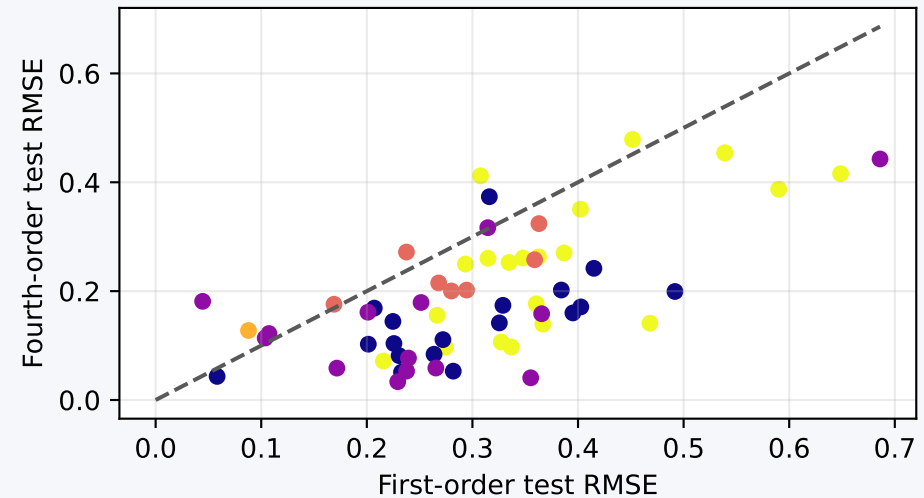
Explicit failure boundary

Coverage only 0.10

Synthetic batch trajectory



Measured late-time extrapolation



Measured: 61 experiments | median fourth-order RMSE 0.171 | first-order 0.308 | coverage 0.10

1 Validate data | 2 Simulate | 3 Calibrate | 4 Hold-out validate | 5 Analyze and optimize safely

Measured benchmark with failed interval coverage, not a predictive, safety, or scale-up signoff